Project: Worked examples
Title: Fillet weld (EN 1993-1-8)
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

A fillet weld carrying forces parallel to, transverse to and normal to its axis, by the simplified method: the resultant over the throat area against the design shear strength of the weld, with the required throat and leg.

WHAT IT CALCULATES	RELATION
Resultant force on the weld	$F = \sqrt{(F_{\parallel}^2 + F_{\perp}^2 + F_n^2)}$
Design shear strength of the weld, simplified method	$f_{vw,d} = f_u/(\sqrt{3}\cdot\beta_w\cdot\gamma_{M2})$
Long-joint reduction of a lap joint longer than 150·a (4.11(3))	$\beta_{Lw} = \min(1, 1.2 - 0.2\cdot L/(150\cdot a))$; 1 for a shorter lap or another joint type
Shear stress on the throat	$\tau_w = F/(a\cdot L)$
Weld resistance and utilisation	$F_{w,Rd} = \beta_{Lw}\cdot f_{vw,d}\cdot a\cdot L$; $u = F/F_{w,Rd}$
Throat needed (not less than 3 mm, with the β_{Lw} of the throat provided) and the matching leg	$a_{req} = \max(F/(\beta_{Lw}\cdot f_{vw,d}\cdot L), 3\ \text{mm})$; leg = $\sqrt{2}\cdot a$

Basis: Simplified method for design resistance of fillet weld[1](4.5.3.3); Effective throat thickness: minimum 3 mm[1](4.5.2(2)); Effective length: welds shorter than 30 mm or 6 throats not to carry load[1](4.5.1(2)); Long joints[1](4.11).

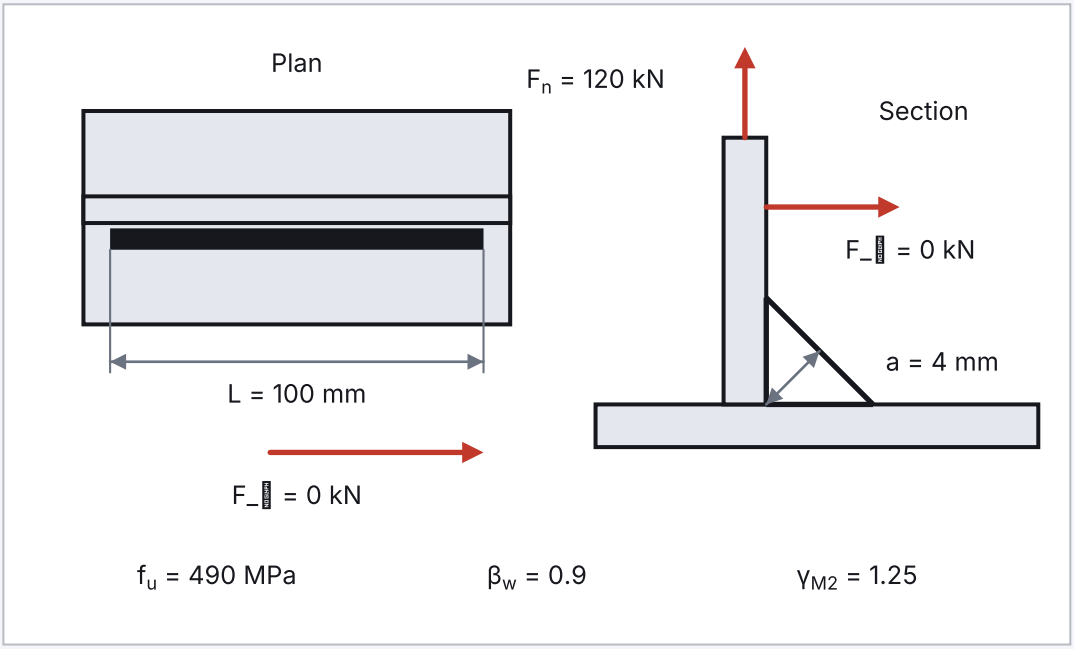


Figure 1 — A fillet weld of effective length L: in plan, the force along it; in section, the throat a and the forces across it and normal to the plate. Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
F_{par}	Force parallel to the weld axis	0	kN	spec: LOAD-SCHEDULE-01	-100000 … 100000 kN
F_{tr}	Force transverse to the weld axis, in the plane of the parts	0	kN	spec: LOAD-SCHEDULE-01	-100000 … 100000 kN
F_n	Force normal to the plane of the parts	120	kN	spec: LOAD-SCHEDULE-01	-100000 … 100000 kN
L	Effective weld length	100	mm	drawing: DRG-WD-702 rev A	30 … 10000 mm
a	Throat thickness	4	mm	drawing: DRG-WD-702 rev A	3 … 50 mm
joint	Joint type for the long-joint reduction of BS EN 1993-1-8 4.11: a lap joint is reduced beyond 150 throats; choose Other for a weld whose stress follows the base metal's, as a flange-to-web weld does	Lap joint		default	any
f_u	Ultimate strength of the weaker part joined	490	MPa	spec: SPEC-STR-02 design criteria	300 … 1000 MPa
β_w	Correlation factor β_w (0.8 S235, 0.85 S275, 0.9 S355, 1.0 S420/S460)	0.9		spec: SPEC-STR-02 design criteria	0.8 … 1
γ_{M2}	Partial factor γ_{M2}	1.25		spec: SPEC-STR-02 design criteria	1 … 2

3. Assumptions

A1. The resultant force is shared uniformly along the effective length. (Simplified method.)
A2. Equal-leg fillet: leg = √2 × throat. (Geometry.)
A3. A lap joint longer than 150·a is reduced by β_{Lw} = 1.2 − 0.2·L/(150·a), not above 1. (BS EN 1993-1-8 4.11(3).)
A4. The weld length L is the length of the lap in the direction of the force. (BS EN 1993-1-8 4.11(3).)

4. Working

[10] β_{Lw} = no long-joint reduction [1](4.11)
= lap not longer than 150·a
= 1
Long-joint reduction factor (1: not reduced)

[11] F = sqrt(F_{par}^2 + F_{tr}^2 + F_n^2)
= sqrt((0 kN)^2 + (0 kN)^2 + (120 kN)^2)
= 120 kN
Resultant force on the weld

[12] $f_{vw,d}$ = f_u / (sqrt(3) * β_w * γ_{M2}) [1](4.5.3.3)
= 490 MPa / (sqrt(3) × 0.9 × 1.25)
= 251.47 MPa
Design shear strength of the weld

[13] $\tau_w = F / (a * L)$
$= 120\ kN / (4\ mm \times 100\ mm)$
$= 300\ MPa$
Stress on the throat

[14] $F_{w,Rd} = \beta_{Lw} * f_{vw,d} * a * L$ [1](4.5.3.3)
$= 1 \times 251.47\ MPa \times 4\ mm \times 100\ mm$
$= 100.59\ kN$
Design resistance of the weld, reduced by the long-joint factor when it applies

[15] $u = F / F_{w,Rd}$
$= 120\ kN / 100.59\ kN$
$= 1.193$
Utilisation

[16] $a_{calc} = F / (\beta_{Lw} * f_{vw,d} * L)$
$= 120\ kN / (1 \times 251.47\ MPa \times 100\ mm)$
$= 4.772\ mm$
Throat needed for the load (with the long-joint factor of the throat provided)

[17] $a_{req} = \max(a_{calc}, 3\ mm)$ [1](4.5.2(2))
$= \max(4.772\ mm, 3\ mm)$
$= 4.772\ mm$
Required throat (not less than 3 mm)

[18] $s_{leg} = \sqrt{2} * a$
$= \sqrt{2} \times 4\ mm$
$= 5.6569\ mm$
Leg length of the throat provided (equal legs)

5. Results

SYMBOL	VALUE	DESCRIPTION
F	120 kN	Resultant force
$f_{vw,d}$	251.5 MPa	Design shear strength
τ_w	300 MPa	Throat stress
β_{Lw}	1	Long-joint factor
$F_{w,Rd}$	100.59 kN	Weld resistance
a_{calc}	4.772 mm	Throat needed
a_{req}	4.772 mm	Required throat
s_{leg}	5.657 mm	Leg length provided
u	1.20	Utilisation (governing)

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
FAIL	Weld resistance	u = 1.20	u ≤ 1	1.20	[1](4.5.3.3)

7. Applicability

None.

Defaults used (inputs left empty):
joint: not entered, so the default of 'Lap joint' is used.

8. Verification

Template struct.fillet-weld@1.1.0 Content hash 05873bcb86c43432 Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. British Standards Institution. Eurocode 3: Design of Steel Structures—Part 1-8: Design of Joints. BS EN 1993-1-8:2005. BSI; 2005.